

OPERATING SYSTEMS

Memory Management

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Memory Management Commands

cat /proc/meminfo – virtual file tells about the amount of memory available and used

free – tells about the physical memory and virtual memory status

top - check memory and CPU usage per process

vmstat – info about processes, memory, paging, block IO, traps, and CPU activity



Procs

r: number of processes waiting for run time

b: number of processes in uninterruptible sleep

Memory

swpd: amount of virtual memory used

free: amount of idle memory

buff: the amount of memory used as buffers

cache: amount of memory used as cache

Swap

si: memory swapped in from disk (/s)

so: memory swapped to disk (/s)

IO

bi: Blocks received from a block device (blocks/s)

bo: Blocks sent to a block device (blocks/s)

System

in: number of interrupts per second, including the clock

cs: number of context switches per second

CPU – These are percentages of total CPU time.

us: Time spent running non-kernel code. (user time, including nice time)

sy: Time spent running kernel code. (system time)

id: Time spent idle.

wa: Time spent waiting for IO

st: Time stolen from a virtual machine

In GUI mode

- Navigate to Show Applications
- Enter System Monitor in the search bar and access the application
- Select the Resources tab
- A graphical overview of your memory consumption in real time, including historical information is displayed



THANK YOU

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